KONGU ENGINEERING COLLEGE

TRANSFORM YOURSELF

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON

ACADEMIC MANAGEMENT SYSTEM

PROJECT REPORT

Submitted by

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25ITR173

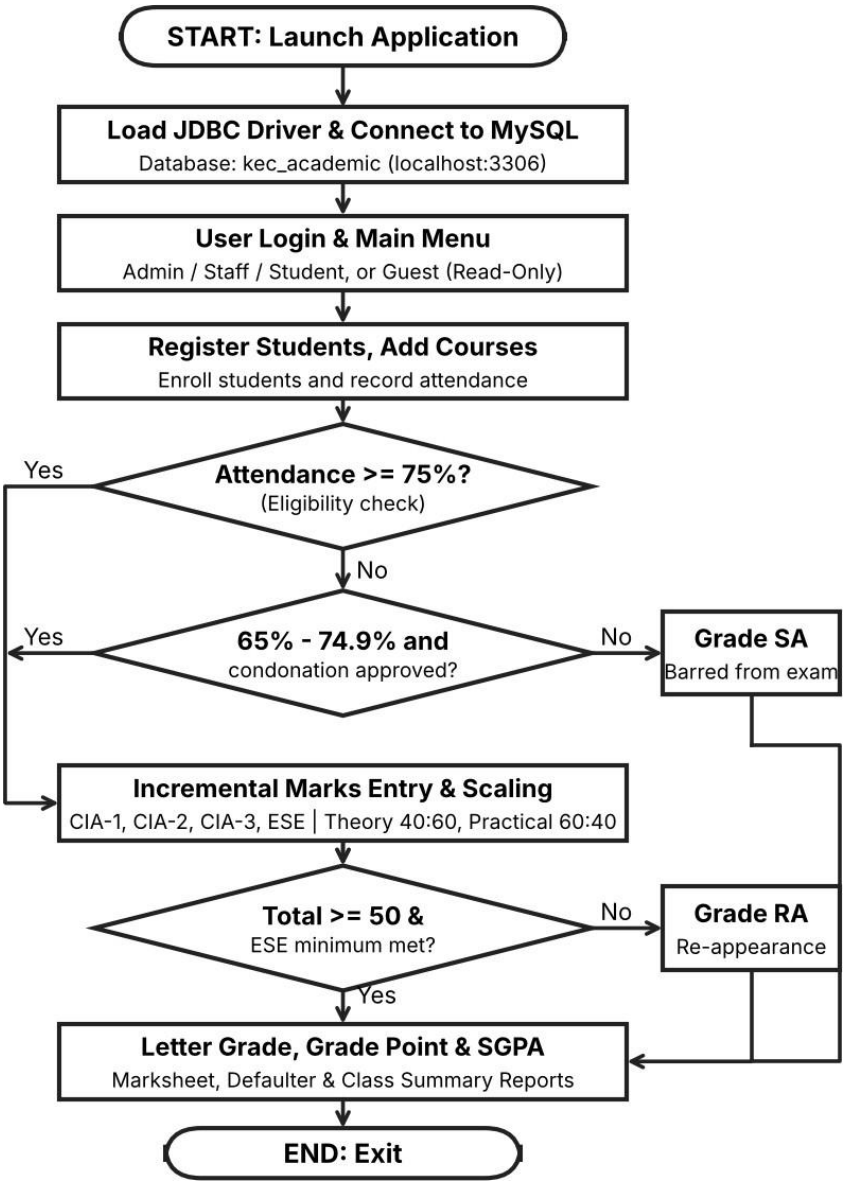
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC and MySQL.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform insert, select, update and delete operations, record incremental marks (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate marksheets and academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver (com.mysql.cj.jdbc.Driver) to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_academic database on the local MySQL server using DriverManager.getConnection(), with the URL, user name and password read from environment variables or default values.

- Authenticate the user against the users table. Administrators and staff get full access, students are restricted, and a guest session has read-only access to the directory.

- Store and retrieve student details (roll number, name, department, semester) and course details (course code, title, credits, course type) using PreparedStatement queries.

- Enroll students in semester courses for an academic year, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance. Students with 75% or above are eligible, those with 65%-74.9% require medical condonation approval, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2 and CIA-3 / assignment marks, and calculate the scaled internal marks according to the course type (Theory 40%, Practical 60%, Embedded 50%).

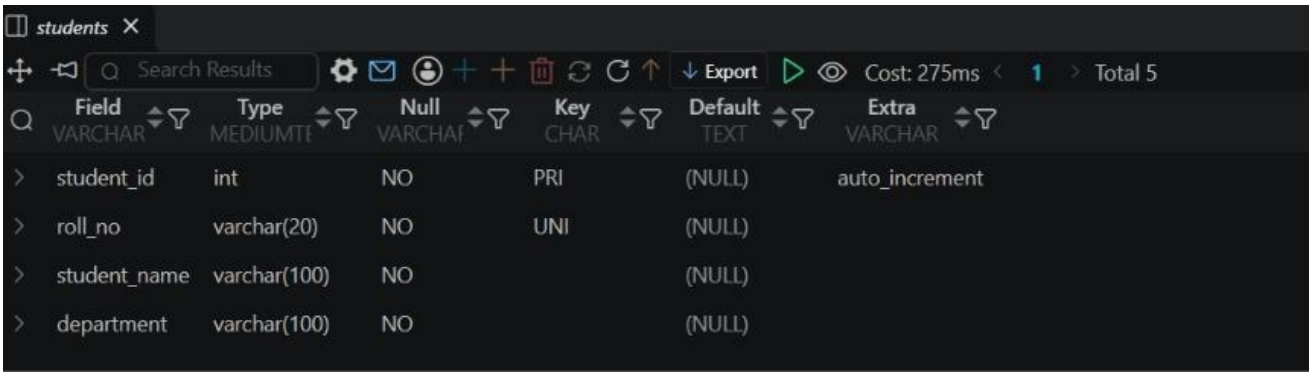
- Record the End Semester Examination (ESE) marks, handle absentees, verify the minimum ESE threshold, and scale the marks according to the course type.

- Calculate the total marks by combining the scaled CIA and ESE marks, assign the letter grade and grade point, and mark the student as RA (Re-Appearance) if the pass requirements are not met.

- Generate the student semester marksheet with SGPA, the attendance defaulter and condonation report, and the class performance and result summary, and display them on the command prompt console.

DATABASE SCHEMA:

1. STUDENTS TABLE:

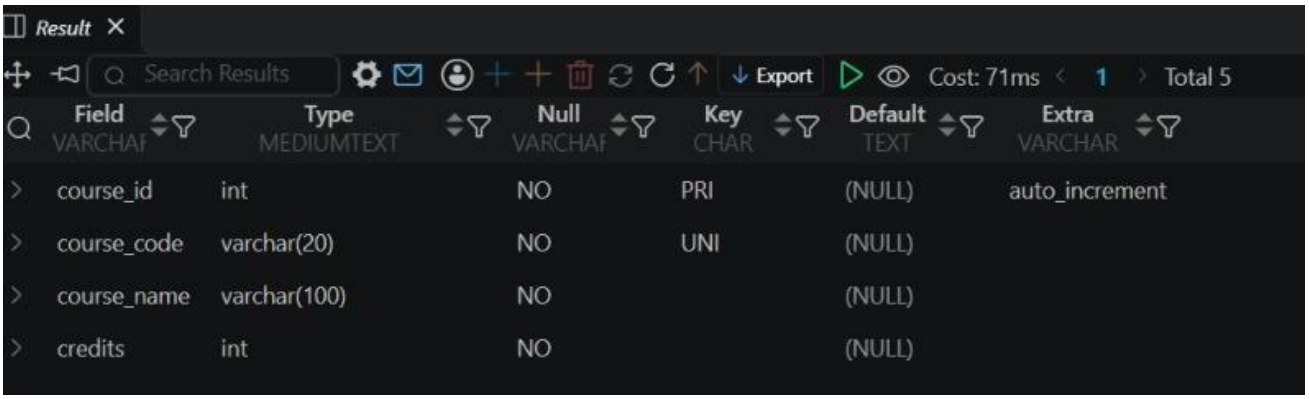


students

Search Results | Export | Cost: 275ms | 1 | Total 5

Field VARCHAR	Type MEDIUMTE	Null VARCHAF	Key CHAR	Default TEXT	Extra VARCHAR
student_id	int	NO	PRI	(NULL)	auto_increment
roll_no	varchar(20)	NO	UNI	(NULL)	
student_name	varchar(100)	NO		(NULL)	
department	varchar(100)	NO		(NULL)	

2. COURSES TABLE:

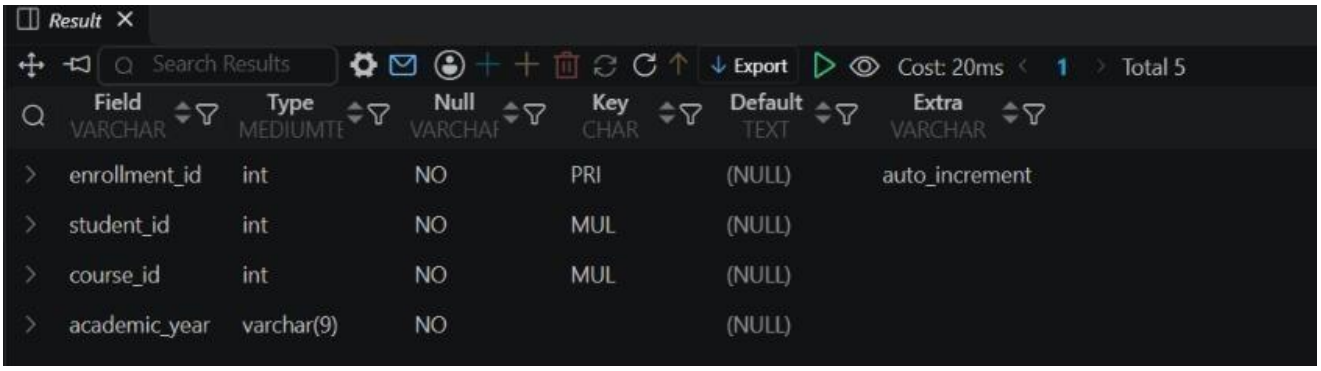


Result

Search Results | Export | Cost: 71ms | 1 | Total 5

Field VARCHAF	Type MEDIUMTEXT	Null VARCHAF	Key CHAR	Default TEXT	Extra VARCHAR
course_id	int	NO	PRI	(NULL)	auto_increment
course_code	varchar(20)	NO	UNI	(NULL)	
course_name	varchar(100)	NO		(NULL)	
credits	int	NO		(NULL)	

3. ENROLLMENTS TABLE:



Result

Search Results | Export | Cost: 20ms | 1 | Total 5

Field VARCHAR	Type MEDIUMTE	Null VARCHAF	Key CHAR	Default TEXT	Extra VARCHAR
enrollment_id	int	NO	PRI	(NULL)	auto_increment
student_id	int	NO	MUL	(NULL)	
course_id	int	NO	MUL	(NULL)	
academic_year	varchar(9)	NO		(NULL)	

4. USERS TABLE:



Result

Search Results | Export | Cost: 20ms | 1 | Total 8

Field VARCHAR	Type MEDIUMTEXT	Null VARCHAF	Key CHAR	Default TEXT	Extra VARCHAR
user_id	int	NO	PRI	(NULL)	auto_increment
username	varchar(50)	NO	UNI	(NULL)	
password_hash	varchar(255)	NO		(NULL)	
role	enum('ADMINISTRATOR','ST	NO	MUL	(NULL)	

SOURCE CODE:

MAIN.JAVA (MENU AND DATABASE OPERATIONS):

```
                case "2" -> addStudent(user);
                case "3" -> listCourses();
                case "4" -> attendance(user);
                case "5" -> marks(user);
                case "0" -> { System.out.println("Goodbye!"); return; }
                default -> System.out.println("Invalid choice.");
            }
        } catch (Exception e) {
            System.out.println("Operation failed: " + e.getMessage());
        }
    }
}
static void listStudents() throws SQLException {
    String sql = "SELECT roll_no, student_name, department, semester " +
                 "FROM students ORDER BY roll_no";
    try (Connection c = db(); Statement s = c.createStatement();
        ResultSet r = s.executeQuery(sql)) {
        System.out.printf("%-14s %-24s %-24s %s%n",
                          "Roll No", "Name", "Department", "Sem");
        while (r.next())
            System.out.printf("%-14s %-24s %-24s %d%n",
                r.getString(1), r.getString(2), r.getString(3), r.getInt(4));
    }
}
static void addStudent(UserSession u) throws SQLException {
    if (u != null && u.isStudent()) {
        System.out.println("Access denied."); return;
    }
    System.out.print("Roll number: ");
    String roll = sc.nextLine().trim().toUpperCase();
    System.out.print("Student name: ");
    String name = sc.nextLine().trim();
    System.out.print("Department: ");
```

```
    String dept = sc.nextLine().trim();
    System.out.print("Semester (1-8): ");
    int sem = Integer.parseInt(sc.nextLine().trim());
    if (roll.isEmpty() || name.isEmpty() || sem < 1 || sem > 8) {
        System.out.println("Invalid student details."); return;
    }
    String sql = "INSERT INTO students(roll_no,student_name,department,semester) " +
                 "VALUES(?,?,?,?)";
    try (Connection c = db(); PreparedStatement p = c.prepareStatement(sql)) {
        p.setString(1, roll); p.setString(2, name);
        p.setString(3, dept); p.setInt(4, sem);
        p.executeUpdate();
        System.out.println("Student registered.");
    }
}
static void listCourses() throws SQLException {
    String sql = "SELECT course_code, course_name, course_type, credits " +
                 "FROM courses ORDER BY course_code";
    try (Connection c = db(); Statement s = c.createStatement();
         ResultSet r = s.executeQuery(sql)) {
        while (r.next())
            System.out.printf("%s | %s | %s | %d credits%n",
                r.getString(1), r.getString(2), r.getString(3), r.getInt(4));
    }
}
static void attendance(UserSession u) throws SQLException {
    if (u != null && u.isStudent()) {
        System.out.println("Access denied."); return;
    }
    System.out.print("Enrollment ID: ");
    int id = Integer.parseInt(sc.nextLine().trim());
    System.out.print("Classes held: ");
    int held = Integer.parseInt(sc.nextLine().trim());
    System.out.print("Classes attended: ");
    int attended = Integer.parseInt(sc.nextLine().trim());
    if (held < 0 || attended < 0 || attended > held) {
```

```
            System.out.println("Invalid attendance values."); return;
        }
        String sql = "INSERT INTO attendance(enrollment_id,classes_held," +
            "classes_attended,condonation_approved) VALUES(?,?,?,0) " +
            "ON DUPLICATE KEY UPDATE classes_held=VALUES(classes_held)," +
            "classes_attended=VALUES(classes_attended)";
        try (Connection c = db(); PreparedStatement p = c.prepareStatement(sql)) {
            p.setInt(1, id); p.setInt(2, held); p.setInt(3, attended);
            p.executeUpdate();
            double pct = held == 0 ? 0 : attended * 100.0 / held;
            System.out.printf("Attendance saved: %.1f%%%n", pct);
        }
    }
    static void marks(UserSession u) throws SQLException {
        if (u != null && u.isStudent()) {
            System.out.println("Access denied."); return;
        }
        System.out.print("Enrollment ID: ");
        int id = Integer.parseInt(sc.nextLine().trim());
        String[] names = {"CIA1", "CIA2", "CIA3", "ESE"};
        String find = "SELECT mark_id FROM assessment_marks " +
                      "WHERE enrollment_id=? AND assessment_name=?";
        String update = "UPDATE assessment_marks SET marks_obtained=? WHERE
        mark_id=?";
        String insert = "INSERT INTO assessment_marks(enrollment_id,assessment_name," +
                        "max_marks,marks_obtained,is_absent) VALUES(?,?,?,?,0)";
        try (Connection c = db()) {
            for (int i = 0; i < names.length; i++) {
                System.out.print(names[i] + " marks: ");
                double mark = Double.parseDouble(sc.nextLine().trim());
                if (mark < 0 || mark > 100) {
                    System.out.println("Marks must be 0-100."); return;
                }
                Integer markId = null;
                try (PreparedStatement p = c.prepareStatement(find)) {
                    p.setInt(1, id); p.setString(2, names[i]);
```

```
                try (ResultSet r = p.executeQuery()) {
                    if (r.next()) markId = r.getInt(1);
                }
            }
            try (PreparedStatement p = c.prepareStatement(
                    markId == null ? insert : update)) {
                if (markId == null) {
                    p.setInt(1, id); p.setString(2, names[i]);
                    p.setDouble(3, 100); p.setDouble(4, mark);
                    p.setBoolean(5, false);
                } else {
                    p.setDouble(1, mark); p.setInt(2, markId);
                }
                p.executeUpdate();
            }
        }
        System.out.println("Marks saved.");
    }
}
```

DATABASECONNECTION.JAVA:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
public class DatabaseConnection {
    // Default connection parameters
    private static final String DEFAULT_DB_URL =
    "jdbc:mysql://127.0.0.1:3306/kec_academic";
    private static final String DEFAULT_DB_USER = "root";
    private static final String DEFAULT_DB_PASSWORD = " ";
    public static String getDbUrl() {
        String url = System.getenv("KEC_DB_URL");
        return (url != null && !url.trim().isEmpty()) ? url.trim() : DEFAULT_DB_URL;
    }
    public static String getDbUser() {
```

```
        String user = System.getenv("KEC_DB_USER");
        return (user != null && !user.trim().isEmpty()) ? user.trim() : DEFAULT_DB_USER;
    }
    public static String getDbPassword() {
        String pass = System.getenv("KEC_DB_PASSWORD");
        return (pass != null) ? pass : DEFAULT_DB_PASSWORD;
    }
    /**
     * Obtains a live JDBC Connection to the MySQL database.
     */
    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(getDbUrl(), getDbUser(), getDbPassword());
    }
}
```

JDBC CONNECTION DETAILS:

Property	**Value**
Database Host	127.0.0.1 (localhost)
Port	3306
Database Name	kec_academic
Username	root
Driver JAR	lib/mysql-connector-j-26.7.0.jar
JDBC Driver Class	com.mysql.cj.jdbc.Driver

OUTPUT SCREENSHOTS:

MAIN MENU:

1. MAIN MENU AND USER AUTHENTICATION:

```
D:\KEC_Java_Project>"C:\Program Files\Java\jdk-21.0.11\bin\javac.exe" -cp "lib/*" -d build src\*.java
Note: src\PdfMarksheetGenerator.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

D:\KEC_Java_Project>"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" -cp "build;lib/*" Main
================================================================================================================
                    KEC ACADEMIX ? KONGU ENGINEERING COLLEGE (AUTONOMOUS)
                     Student Academic & Continuous Evaluation System (R2024)
================================================================================================================

--- User Authentication ---
Please sign in with your KEC credentials, or enter 'guest' for read-only directory access.
Username: guest
Continuing with Guest / Public access (Read-Only).

================================================================================================================
                                        KEC ACADEMIX ? MAIN MENU
   [Active Session: Guest / Public (Read-Only Access)]
================================================================================================================
  1. Student Registration (Directory, Add, Edit Profile)
  2. Course Catalog Management (Curriculum, Types, Credits)
  3. Course Enrollment & Attendance Tracking (Allocations & Attendance %)
  4. Incremental Marks Entry (CIA1, CIA2, CIA3, ESE & Absentee Flags)
  5. Student Semester Marksheet (R2024 Evaluation, SGPA & PDF Export)
  6. Attendance Defaulter & Condonation Report (65%-75% Approval Workflow)
  7. Class Performance & Result Summary (Academic Analytics & Pass %)
  8. Automated Demonstration / Proof of Execution (R2024 Rules Engine Test)
  9. Web Dashboard & Swing Application Status
  0. Exit Console
================================================================================================================
Enter your choice (0-9, or 'L' to switch user):
```

STUDENT & COURSE MANAGEMENT:

1. STUDENT REGISTRATION:

```
                        1. STUDENT REGISTRATION MANAGEMENT
--------------------------------------------------------------------------------
  1. List All Registered Students
  2. Search Student by Roll Number
  3. Register New Student (Admin / Staff)
  4. Edit Student Information (Admin / Staff)
  0. Back to Main Menu
--------------------------------------------------------------------------------
Choose an option: 1

================================================================================
                              REGISTERED STUDENTS
================================================================================
ID    Roll Number     Full Name                Department             Semester
--------------------------------------------------------------------------------
1     24CSE001        Arun Kumar               CSE                    4
2     24CSE002        Priya Devi               CSE                    4
3     24IT001         Karthik Raj              IT                     4
6     25ITR161        Thina Rajkumar R         Information Technology 4
4     25ITR173        VIKNESHWARAN N           IT                     4
================================================================================
```

2. COURSE CATALOG:

```
--------------------------------------------------------------------------------
                      2. COURSE CATALOG MANAGEMENT
--------------------------------------------------------------------------------
  1. List All Curriculum Courses
  2. Add New Course (Admin / Staff)
  3. Edit Course Information (Admin / Staff)
  4. Delete Course (Protected against active enrollments)
  0. Back to Main Menu
--------------------------------------------------------------------------------
Choose an option: 2

--- Add New Curriculum Course ---
Enter Course Code (e.g. CS2403): IT3448
Enter Course Name: DBMS
Select Course Type:
  1. THEORY (40% CIA / 60% ESE, Min ESE 27)
  2. PRACTICAL (60% CIA / 40% ESE, Min ESE 20)
  3. EMBEDDED (50% CIA / 50% ESE, Min ESE 22.5)
Choice (1-3) [1]: 1
Enter Credits (1 to 10) [3]: 4

>>> Course 'IT3448 - DBMS' [THEORY, 4 Credits] added successfully. <<<
```

3. COURSE ENROLLMENT:

```
                3. COURSE ENROLLMENT & ATTENDANCE TRACKING
 ------------------------------------------------------------------------------
  1. View Active Enrollments & Attendance Status
  2. Enroll Student in Course (Admin / Staff)
  3. Record / Update Attendance (Admin / Staff)
  0. Back to Main Menu
 ------------------------------------------------------------------------------
Choose an option: 2

--- Enroll Student in Curriculum Course ---
Enter Student Roll Number: 25ITR173
Enter Course Code: IT3448
Enter Academic Year (e.g. 2026-2027) [2026-2027]: 2026-2027
Enter Semester (1-8) [4]: 3

>>> Successfully enrolled VIKNESHWARAN N (25ITR173) in IT3448 (DBMS) for Sem 3, 2026-2027. <<<
```

4. ATTENDANCE TRACKING:

```
--------------------------------------------------------------------------------
                3. COURSE ENROLLMENT & ATTENDANCE TRACKING
--------------------------------------------------------------------------------
  1. View Active Enrollments & Attendance Status
  2. Enroll Student in Course (Admin / Staff)
  3. Record / Update Attendance (Admin / Staff)
  0. Back to Main Menu
--------------------------------------------------------------------------------
Choose an option: 3
Enter Enrollment ID (or press Enter to search by Roll & Course):
Enter Student Roll Number: 25ITR173
Enter Course Code: IT3448

Updating Attendance for: VIKNESHWARAN N (25ITR173) ? Course: IT3448 (DBMS)
Current Record: 0 Classes Held, 0 Classes Attended
Enter Total Classes Held [0]: 1
Enter Classes Attended [0]: 1
Calculated Attendance: 100.0% ? Status: Eligible

>>> Attendance recorded successfully. <<<
```

INCREMENTAL MARKS ENTRY:

1. CIA AND ESE MARKS ENTRY (R2024):

```
--------------------------------------------------------------------------------
                        4. INCREMENTAL MARKS ENTRY (R2024)
--------------------------------------------------------------------------------
Enter Student Roll Number: 25ITR173
Enter Course Code: IT3448

================================================================================
Student : VIKNESHWARAN N (25ITR173)
Course  : IT3448 - DBMS [Theory, 4 Credits]
Attendance : 100.0% (Eligible)
================================================================================
Enter Continuous Internal Assessment Marks (0 - 100):
  CIA 1 (out of 100) [0.0]: 34
  CIA 2 (out of 100) [0.0]: 34
  CIA 3 / Assignment / Quiz (out of 100) [0.0]: 25
Was the student ABSENT in End Semester Exam (ESE)? (Y/N) [N]:
  End Semester Exam (out of 60, Min threshold 27.0) [0.0]: 54

--------------------------------------------------------------------------------
                             EVALUATION PREVIEW
--------------------------------------------------------------------------------
Scaled Internal : 12.40 / 40 (Scale: 40%)
End Semester    : 54.00 / 60 (Min required: 27.0)
Total Aggregate : 66.40 / 100
Letter Grade    : B+ (Grade Point: 7)
Academic Result : PASS
--------------------------------------------------------------------------------
Save these marks to the database? (Y/N): Y

>>> All assessment marks saved successfully into database. <<<
```

ATTENDANCE & GRADING RULES:

1. ATTENDANCE DEFAULTER AND CONDONATION REPORT:

```
--------------------------------------------------------------------------
            6. ATTENDANCE DEFAULTER & CONDONATION REPORT (R2024)
--------------------------------------------------------------------------
  1. View All Attendance Records & Defaulters (<75%)
  2. View Condonation Candidates (65.0% - 74.9%)
  3. View Critical Attendance Shortage / Detained (<65% SA)
  4. Approve Medical Condonation (Admin / Staff)
  0. Back to Main Menu
--------------------------------------------------------------------------
Choose an option: 1

================================================================================================================================
                                        ATTENDANCE DEFAULTER & ELIGIBILITY AUDIT
================================================================================================================================
AttID Roll No      Student Name         Course     Course Title           Held     Attended Att %      Eligibility Status
--------------------------------------------------------------------------------------------------------------------------------
1     24CSE001     Arun Kumar           CS2401     Java Programming       45       40       88.9      % Eligible
4     24CSE001     Arun Kumar           CS2402     Database Managemen...  45       40       88.9      % Eligible
2     24CSE002     Priya Devi           CS2401     Java Programming       45       40       88.9      % Eligible
5     24CSE002     Priya Devi           CS2402     Database Managemen...  45       40       88.9      % Eligible
3     24IT001      Karthik Raj          CS2401     Java Programming       45       40       88.9      % Eligible
14    25ITR173     VIKNESHWARAN N       IT3448     DBMS                   1        1        100.0     % Eligible
--------------------------------------------------------------------------------------------------------------------------------
Total Records: 6 | Fully Eligible: 6 | Condonation Required (65-74.9%): 0 | Detained (SA <65%): 0
================================================================================================================================

--------------------------------------------------------------------------
            6. ATTENDANCE DEFAULTER & CONDONATION REPORT (R2024)
--------------------------------------------------------------------------
  1. View All Attendance Records & Defaulters (<75%)
  2. View Condonation Candidates (65.0% - 74.9%)
  3. View Critical Attendance Shortage / Detained (<65% SA)
  4. Approve Medical Condonation (Admin / Staff)
  0. Back to Main Menu
--------------------------------------------------------------------------
Choose an option:
```

2. STUDENT SEMESTER MARKSHEET:

```
5. STUDENT SEMESTER MARKSHEET (R2024)

Enter student roll number: 25ITR173
Enter semester (1-8, or press Enter for all): 3

KEC ACADEMIX ? STUDENT ACADEMIC MARKS REPORT (R2024)

Student Name  : VIKNESHWARAN N          Roll Number   : 25ITR173
Department    : IT                     Academic Year : 2026-2027
Semester      : 4                      Filter        : Semester 3

Code       Course Name                Type       Credits CIA          ESE          Total      Grade  GP   Result         Attendance
IT3448     DBMS                       Theory     4       12.40/40     54.00/60     66.40      B+     7    PASS           100.0%

EVALUATION SUMMARY

Total Credits Registered : 4
Total Credits Earned     : 4
Semester SGPA            : 7.00
Equivalent Percentage    : 70.00%
Academic Status          : PASS
```

3. CLASS PERFORMANCE AND RESULT SUMMARY:

```
7. CLASS PERFORMANCE & RESULT SUMMARY (R2024)

Enter Department filter (or press Enter for all):
Enter Semester filter (1-8, or press Enter for all): 3
Enter Academic Year filter (e.g. 2026-2027, or press Enter for all): 2026-2027

INSTITUTIONAL ACADEMIC PERFORMANCE SUMMARY

Total Distinct Enrolled Students : 1
Students Evaluated (Complete)    : 1
Incomplete Evaluations (Pending) : 0

Passed All Registered Courses    : 1
Reappear (RA) / Failed Courses   : 0
  - Shortage of Attendance (SA)  : 0
  - Absent in Exam (AB)          : 0

Overall Cohort Pass Percentage   : 100.00%
Cohort Average SGPA              : 7.00
```

4. AUTOMATED DEMONSTRATION / PROOF OF EXECUTION:

```
8. AUTOMATED DEMONSTRATION & PROOF OF EXECUTION (R2024 ENGINE)

Running safe, in-memory validation of authoritative academic formulas without modifying database records...

  [PASS] 1. Theory Course (40/60 split, CIA 80->32, ESE 35->35, Total 67, Grade B+)
  [PASS] 2. Theory ESE Minimum Failure (ESE 26.5 < 27 yields RA, GP 0 regardless of internal 40/40)
  [PASS] 3. Practical Course (60/40 split, CIA 90->54, ESE 20->20, Total 74, Grade A)
  [PASS] 4. Embedded Course (50/50 split, CIA 80->40, ESE 22.5->22.5, Total 62.5, Grade B+)
  [PASS] 5. Attendance Precedence Rule (<65% attendance detains student as SA despite 100% marks)
  [PASS] 6. Medical Condonation Logic (70% unapproved is ineligible, approved condonation becomes eligible)
  [PASS] 7. Credit-Weighted SGPA Calculation (SGPA: 7.22, Equivalent Percentage: 72.20%)
  [PASS] 8. Live Database Connection & Record Safety Verified (5 live student records)

Demonstration Summary: 8 / 8 Tests Passed Successfully (100% Integrity)
```

DATABASE:

STUDENTS, COURSES AND ENROLLMENTS TABLES (MySQL):

```
mysql> SELECT * FROM students;
+------------+----------+------------------+-----------------------+----------+
| student_id | roll_no  | student_name     | department            | semester |
+------------+----------+------------------+-----------------------+----------+
|          1 | 24CSE001 | Arun Kumar       | CSE                   |        4 |
|          2 | 24CSE002 | Priya Devi       | CSE                   |        4 |
|          3 | 24IT001  | Karthik Raj      | IT                    |        4 |
|          4 | 25ITR173 | VIKNESHWARAN N   | IT                    |        4 |
|          6 | 25ITR161 | Thina Rajkumar R | Information Technology |        4 |
+------------+----------+------------------+-----------------------+----------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM courses;
+-----------+-------------+-----------------------------+---------+-------------+
| course_id | course_code | course_name                 | credits | course_type |
+-----------+-------------+-----------------------------+---------+-------------+
|         1 | CS2401      | Java Programming            |       4 | THEORY      |
|         2 | CS2402      | Database Management Systems |       3 | THEORY      |
|        15 | IT3448      | DBMS                        |       4 | THEORY      |
+-----------+-------------+-----------------------------+---------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM enrollments;
+---------------+------------+-----------+---------------+----------+
| enrollment_id | student_id | course_id | academic_year | semester |
+---------------+------------+-----------+---------------+----------+
|             1 |          1 |         1 | 2026-2027     |        4 |
|             2 |          1 |         2 | 2026-2027     |        4 |
|             3 |          2 |         1 | 2026-2027     |        4 |
|             4 |          2 |         2 | 2026-2027     |        4 |
|             5 |          3 |         1 | 2026-2027     |        4 |
|            18 |          4 |        15 | 2026-2027     |        3 |
+---------------+------------+-----------+---------------+----------+
6 rows in set (0.00 sec)

mysql>
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The application connected to the database successfully and automates attendance eligibility, Continuous Internal Assessment (CIA) and End Semester Examination (ESE) marks, letter grades and the Semester Grade Point Average (SGPA).

REFERENCES:

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2. Oracle, Java SE API – HttpServer. https://docs.oracle.com/en/java/javase/17/docs/api/jdk.httpserver/com/sun/net/httpserver/HttpServer.html
3. Oracle, MySQL 8.0 Reference Manual. https://dev.mysql.com/doc/refman/8.0/en/
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GITHUB:

LINK:	**https://github.com/VikneshwaranNv/Java_JDBC_project**